

Azure Data Engineering Interview Preparation

Beginner → Intermediate → Advanced → 4–5 Year Interview Level

Purpose: Use this as a printable daily study workbook. Learn one level at a time, understand the answers in your own words, and connect each concept to hands-on Azure practice.

Level	Focus	Questions
1	Foundation / Basics	36
2	Intermediate Azure Data Engineering	40
3	Advanced / 4–5 Year Level	40
4	SQL Interview Preparation	40
5	Databricks + PySpark	38
6	Project + Scenario Interview	36

Recommended daily routine: 4–6 questions per day, write short notes, practice the concept, and answer aloud without reading. Revise yesterday's questions before starting new ones.

LEVEL 1 — FOUNDATION / BASIC Q&A;

Question count: 36

Q1. What is Data Engineering?

Answer: Data Engineering is the process of collecting, storing, transforming, and preparing data so it can be used for analytics and reporting.

Q2. What is ETL?

Answer: ETL means Extract, Transform, Load. Data is extracted from a source, transformed, and then loaded into the target.

Q3. What is ELT?

Answer: ELT means Extract, Load, Transform. Data is first loaded into a platform such as a data lake or warehouse and transformed there.

Q4. ETL vs ELT?

Answer: ETL transforms before loading; ELT loads first and transforms later. Cloud platforms commonly support ELT well.

Q5. What is Microsoft Azure?

Answer: Azure is Microsoft's cloud platform providing compute, storage, databases, integration, analytics, security, and many other services.

Q6. What is Azure Data Factory?

Answer: ADF is a cloud data integration and orchestration service used to move data and coordinate data-processing workflows.

Q7. What is an ADF pipeline?

Answer: A pipeline is a logical workflow containing activities that together perform a data integration task.

Q8. What is an ADF activity?

Answer: An activity is one step in a pipeline, such as Copy Data, Lookup, Get Metadata, ForEach, or Execute Pipeline.

Q9. What is a dataset?

Answer: A dataset represents the data structure or location that an ADF activity reads from or writes to.

Q10. What is a linked service?

Answer: A linked service contains connection information used by ADF to connect to a data store or compute service.

Q11. Linked Service vs Dataset?

Answer: Linked Service answers 'how/where do I connect?'; Dataset answers 'what data am I using?'

Q12. What is Integration Runtime?

Answer: Integration Runtime provides the compute and connectivity infrastructure used by ADF for data movement and activities.

Q13. What is ADLS Gen2?

Answer: Azure Data Lake Storage Gen2 is Azure storage designed for large-scale analytics data and supports hierarchical folders and access control.

Q14. What is a storage account?

Answer: A storage account provides the namespace and configuration for Azure Storage services.

Q15. What is a container?

Answer: A container is a top-level logical filesystem boundary in Blob Storage/ADLS Gen2.

Q16. What is a data lake?

Answer: A data lake is centralized storage for large volumes of structured, semi-structured, and unstructured data.

Q17. What is a data warehouse?

Answer: A data warehouse is an analytical system optimized for structured data, reporting, and business intelligence.

Q18. Data Lake vs Data Warehouse?

Answer: A lake is flexible and can store raw data of many types; a warehouse is structured and optimized for analytical queries.

Q19. What is Copy Data activity?

Answer: Copy Data moves data from a source to a sink and is one of the most common ADF activities.

Q20. What is Get Metadata?

Answer: Get Metadata retrieves information such as file names, size, modified time, existence, or child items.

Q21. What is ForEach?

Answer: ForEach iterates over a collection, allowing the same activity or group of activities to run for each item.

Q22. What is a parameter in ADF?

Answer: A parameter is an input value that helps make a pipeline or dataset reusable and dynamic.

Q23. What is a variable in ADF?

Answer: A variable is a value that can be changed during pipeline execution and used for control logic or temporary state.

Q24. What is dynamic content?

Answer: Dynamic content uses ADF expressions to calculate or retrieve values at runtime, for example @item().name.

Q25. What is a trigger?

Answer: A trigger starts a pipeline automatically based on a schedule, event, or time window.

Q26. What is full load?

Answer: Full load copies all required source data to the target, commonly for an initial load or small datasets.

Q27. What is incremental load?

Answer: Incremental load processes only new or changed data, reducing processing time and cost.

Q28. What is error handling in ADF?

Answer: It means designing activities and dependencies so failures can be captured, retried, logged, or routed to an alert path.

Q29. What is monitoring in ADF?

Answer: Monitoring shows pipeline and activity runs, status, duration, errors, and execution details.

Q30. What is Git integration in ADF?

Answer: Git integration provides source control for ADF development, enabling branches, version history, collaboration, and pull requests.

Q31. What is Azure Key Vault?

Answer: Key Vault securely stores secrets, keys, and certificates so sensitive values do not have to be hardcoded.

Q32. What is Managed Identity?

Answer: Managed Identity lets Azure resources authenticate to other Azure services without storing credentials in code or configuration.

Q33. What is RBAC?

Answer: Role-Based Access Control determines who can perform which actions on Azure resources.

Q34. What is a raw layer?

Answer: The raw layer preserves source data with minimal transformation and provides a reliable landing area.

Q35. What is a processed layer?

Answer: The processed layer contains data after cleaning, standardization, validation, or other transformations.

Q36. Explain a simple Azure pipeline.

Answer: A common flow is Source → ADF → ADLS Raw → Transformation → Processed/Curated → Warehouse → BI.

Revision checklist: ■ I understand the concept ■ I can explain it without notes ■ I can give a practical example ■ I can answer follow-up questions

LEVEL 2 — INTERMEDIATE AZURE DATA ENGINEERING

Question count: 39

Q1. Why use parameters in ADF?

Answer: Parameters reduce hardcoding and allow the same pipeline or dataset to process different files, tables, dates, or environments.

Q2. Pipeline parameter vs dataset parameter?

Answer: A pipeline parameter belongs to the pipeline; a dataset parameter is defined in the dataset and receives a value from the pipeline/activity.

Q3. Why use @item().name in ForEach?

Answer: It returns the current item's name, which is useful for dynamically building file paths or filenames.

Q4. What is Lookup activity?

Answer: Lookup reads a small set of data or configuration from a source and makes the result available to later activities.

Q5. What is If Condition?

Answer: It evaluates an expression and runs one branch when true and another when false.

Q6. What is Until activity?

Answer: Until repeatedly executes activities until a specified expression becomes true.

Q7. What is Filter activity?

Answer: Filter reduces an input array to only the items that satisfy a condition.

Q8. What is Execute Pipeline?

Answer: It invokes another pipeline, allowing workflows to be modular and reusable.

Q9. What are dependency conditions?

Answer: They control when a downstream activity runs: after success, failure, completion, or skip.

Q10. Why configure retry?

Answer: Retry can recover from temporary failures such as transient network or service issues without manual intervention.

Q11. What is a schedule trigger?

Answer: It starts a pipeline at defined times or recurring intervals.

Q12. What is an event trigger?

Answer: It starts a pipeline when a supported event occurs, such as a file arriving in storage.

Q13. What is a tumbling window trigger?

Answer: It processes fixed, contiguous time windows and is useful for time-based incremental processing.

Q14. What is a watermark?

Answer: A watermark is a stored value such as a timestamp or increasing ID that identifies the point up to which data has been processed.

Q15. How does incremental loading work?

Answer: Read the last successful watermark, extract records greater than it, load them, and update the watermark only after successful processing.

Q16. Why should watermark update happen after successful load?

Answer: Otherwise a failed load could advance the watermark and cause records to be skipped on the next run.

Q17. What is idempotency?

Answer: An idempotent pipeline produces the same correct result when the same input is processed again, avoiding duplicate or inconsistent output.

Q18. How do you prevent duplicate file processing?

Answer: Use metadata/control tables, processed-file tracking, unique keys, checkpoints, or move/archive files after successful processing.

Q19. Why use Parquet?

Answer: Parquet is a columnar format that compresses well and is efficient for analytical workloads.

Q20. CSV vs Parquet?

Answer: CSV is simple and portable but text-based and less efficient; Parquet is columnar, typed, compressed, and usually better for analytics.

Q21. What is partitioning?

Answer: Partitioning divides data into logical groups, often by date or another high-value column, to improve query and processing efficiency.

Q22. What is a metadata-driven pipeline?

Answer: A generic pipeline reads configuration such as source, target, load type, and query from metadata instead of hardcoding each dataset.

Q23. Why use metadata-driven design?

Answer: It reduces duplicated pipelines, improves maintainability, and makes onboarding new tables or files easier.

Q24. How do you process multiple files dynamically?

Answer: Use Get Metadata to obtain childItems, pass the array to ForEach, and use @item().name or parameters inside Copy Data.

Q25. How do you pass a filename to a parameterized dataset?

Answer: Define a dataset parameter such as FileName and pass a dynamic value such as @item().name from the pipeline.

Q26. How do you handle a failed file in a batch?

Answer: Log the file and error, isolate failures where possible, retry transient errors, and route failed files for reprocessing.

Q27. How do you improve ADF performance?

Answer: Use appropriate parallelism, avoid unnecessary activities, optimize source queries, partition large transfers, and choose suitable Integration Runtime resources.

Q28. What is parallel copy?

Answer: Parallel copy allows ADF to transfer data concurrently, potentially reducing elapsed time for large workloads.

Q29. What is schema drift?

Answer: Schema drift means the incoming data structure changes, such as added or removed columns, without a fixed schema being guaranteed.

Q30. How should schema changes be handled?

Answer: Detect and validate changes, make transformation logic tolerant where appropriate, and establish a process for controlled schema evolution.

Q31. What is data validation?

Answer: Validation checks whether incoming data meets expected rules such as required fields, types, ranges, uniqueness, and record counts.

Q32. Why separate raw and curated data?

Answer: It preserves original data for audit/reprocessing while providing clean, business-ready data for consumers.

Q33. What is a control table?

Answer: A control table stores pipeline metadata such as last successful load time, source/target information, status, and batch identifiers.

Q34. What is a batch ID?

Answer: A batch ID uniquely identifies one ingestion or processing run, helping with auditing, troubleshooting, and reconciliation.

Q35. What is a dead-letter or quarantine area?

Answer: It is a location where invalid or failed records/files are isolated instead of blocking all valid processing.

Q36. What is a reusable child pipeline?

Answer: A child pipeline contains common logic that can be called by multiple parent pipelines using parameters.

Q37. Why use separate Dev/Test/Prod environments?

Answer: They isolate development changes from testing and production workloads and reduce deployment risk.

Q38. What is CI/CD for ADF?

Answer: CI/CD automates validation, versioned deployment, and promotion of data factory changes between environments.

Q39. What should you monitor in production?

Answer: Pipeline status, duration, data volumes, failures, retries, SLA delays, source availability, and data-quality results.

Revision checklist: ■ I understand the concept ■ I can explain it without notes ■ I can give a practical example ■ I can answer follow-up questions

LEVEL 3 — ADVANCED / 4–5 YEAR INTERVIEW LEVEL

Question count: 38

Q1. Explain SCD Type 1.

Answer: Type 1 overwrites the existing value and does not preserve historical versions.

Q2. Explain SCD Type 2.

Answer: Type 2 preserves history by creating a new version of a record, commonly using effective dates and a current-row flag.

Q3. What is CDC?

Answer: Change Data Capture identifies inserts, updates, and deletes made in a source so only changes need to be processed.

Q4. Full load vs incremental load in production?

Answer: Use full load when appropriate for initial or small datasets; use incremental/CDC for scalable recurring loads.

Q5. What is the medallion architecture?

Answer: Bronze stores raw data, Silver stores cleaned/conformed data, and Gold stores business-ready data.

Q6. Why use Bronze/Silver/Gold?

Answer: It separates ingestion, transformation, and consumption concerns and improves lineage, reprocessing, and governance.

Q7. What is Delta Lake?

Answer: Delta Lake adds transactional and reliability features such as ACID transactions, schema enforcement/evolution, and MERGE support to data lake storage.

Q8. What is MERGE?

Answer: MERGE can insert, update, or delete target records based on matching source and target conditions, commonly used for upserts.

Q9. What is ACID?

Answer: ACID means Atomicity, Consistency, Isolation, and Durability, providing reliable transactional behavior.

Q10. What is schema enforcement?

Answer: It prevents incompatible data from being written when the target schema does not match expected rules.

Q11. What is schema evolution?

Answer: It allows controlled changes to a dataset schema, such as adding columns, when the platform and process support it.

Q12. What is late-arriving data?

Answer: Data that arrives after the expected processing window; it requires logic to update affected historical partitions or dimensions.

Q13. What is a slowly changing dimension?

Answer: A dimension whose attribute values change over time, requiring a strategy to overwrite or preserve history.

Q14. How do you design an incremental pipeline?

Answer: Identify a reliable change column/CDC source, store the last successful checkpoint, extract changes, validate/load them, then advance the checkpoint.

Q15. How do you make a pipeline restartable?

Answer: Use checkpoints, batch IDs, idempotent writes, clear status tracking, and separate completed work from pending work.

Q16. How do you handle partial failure?

Answer: Track each unit of work, make writes idempotent, isolate failed units, and restart from the last safe checkpoint.

Q17. How do you optimize a slow pipeline?

Answer: Find the bottleneck first, then optimize source queries, file sizes, partitioning, parallelism, transformations, and unnecessary data movement.

Q18. What is small-file problem?

Answer: Too many tiny files increase metadata and task overhead and can reduce performance in data-lake processing.

Q19. How do you address small files?

Answer: Compact files into larger appropriate sizes and avoid generating excessive tiny output partitions.

Q20. What is data skew?

Answer: Data skew occurs when some partitions contain far more records than others, causing uneven task workloads.

Q21. How do you handle skew?

Answer: Identify skewed keys, repartition appropriately, use salting where justified, and optimize joins/aggregations.

Q22. What is broadcast join?

Answer: A broadcast join sends a small dataset to worker nodes so it can be joined locally with a large dataset, reducing shuffle when appropriate.

Q23. What is shuffle?

Answer: Shuffle redistributes data across workers, commonly for joins and aggregations, and can be expensive.

Q24. What is partition pruning?

Answer: The engine reads only relevant partitions instead of scanning all data, reducing I/O.

Q25. What is predicate pushdown?

Answer: Filters are pushed closer to the data source so unnecessary rows are not read or processed.

Q26. How do you secure ADLS?

Answer: Use RBAC/ACLs, managed identities, private networking where required, encryption, least privilege, and secure secret management.

Q27. RBAC vs ACL in ADLS?

Answer: RBAC grants permissions at Azure resource scope; ACLs provide fine-grained filesystem/path permissions in hierarchical storage.

Q28. Why use Managed Identity instead of secrets?

Answer: It removes the need to store and rotate application credentials and integrates with Azure identity and access control.

Q29. What is Key Vault integration?

Answer: ADF or other services can retrieve secrets securely from Key Vault rather than storing credentials directly in pipeline definitions.

Q30. How do you design monitoring?

Answer: Capture run status, durations, row/file counts, errors, batch IDs, and data-quality results, and send alerts for failures or SLA breaches.

Q31. What is data lineage?

Answer: Lineage shows where data came from, how it was transformed, and where it is consumed.

Q32. What is reconciliation?

Answer: Reconciliation compares source and target counts, totals, keys, or other metrics to confirm data was transferred correctly.

Q33. What is SLA?

Answer: A Service Level Agreement defines expected service or delivery performance, such as a pipeline completing by a specific time.

Q34. How do you reduce cloud cost?

Answer: Process only necessary data, use incremental loads, right-size compute, control parallelism, compact files, and shut down unused resources.

Q35. What is environment parameterization?

Answer: It lets the same deployment artifact use different values for Dev, Test, and Prod, such as storage paths or database names.

Q36. How do you promote ADF changes?

Answer: Develop in source control, validate and review changes, deploy through a controlled CI/CD process, and verify the target environment.

Q37. What makes a production pipeline robust?

Answer: It is parameterized, monitored, secure, restartable, idempotent, scalable, auditable, and has clear failure/recovery paths.

Q38. How would you explain a 4–5 year engineer's contribution?

Answer: Focus on design decisions, reusable frameworks, performance, reliability, security, deployment, monitoring, and production problem solving—not just activity configuration.

Revision checklist: ■ I understand the concept ■ I can explain it without notes ■ I can give a practical example ■ I can answer follow-up questions

LEVEL 4 — SQL INTERVIEW PREPARATION

Question count: 38

Q1. WHERE vs HAVING?

Answer: WHERE filters rows before grouping; HAVING filters groups after GROUP BY.

Q2. INNER JOIN vs LEFT JOIN?

Answer: INNER JOIN returns matching rows from both tables; LEFT JOIN keeps every row from the left table and matching rows from the right.

Q3. What is a CTE?

Answer: A Common Table Expression is a named temporary query result defined with WITH and used by the following statement.

Q4. Why use CTEs?

Answer: They improve readability and allow complex logic to be broken into understandable steps.

Q5. ROW_NUMBER vs RANK vs DENSE_RANK?

Answer: ROW_NUMBER gives unique sequential numbers; RANK leaves gaps after ties; DENSE_RANK does not leave gaps after ties.

Q6. How do you find duplicates?

Answer: Group by the business key and use HAVING COUNT(*) > 1, or use ROW_NUMBER to identify extra rows.

Q7. How do you get the second-highest salary?

Answer: Use a ranking function such as DENSE_RANK over salary descending and select rank 2, or use an appropriate subquery.

Q8. What is a window function?

Answer: It calculates across related rows without collapsing them into one row per group.

Q9. What does LAG do?

Answer: LAG returns a value from a previous row within a defined window.

Q10. What does LEAD do?

Answer: LEAD returns a value from a following row within a defined window.

Q11. How do you calculate a running total?

Answer: Use SUM(column) OVER (ORDER BY date/ID) with an appropriate window frame.

Q12. GROUP BY vs window function?

Answer: GROUP BY reduces rows into groups; window functions keep the original rows while adding calculations.

Q13. What is normalization?

Answer: Normalization organizes relational data to reduce redundancy and update anomalies.

Q14. What is denormalization?

Answer: Denormalization intentionally introduces redundancy to improve read performance or simplify analytical access.

Q15. UNION vs UNION ALL?

Answer: UNION removes duplicates; UNION ALL combines results without duplicate elimination and is generally faster.

Q16. What is an index?

Answer: An index is a data structure that can speed up lookups, joins, and sorting, at the cost of storage and write overhead.

Q17. Why can a query be slow?

Answer: Possible causes include poor joins, missing/ineffective indexes, large scans, bad predicates, skew, blocking, stale statistics, or excessive data movement.

Q18. How do you troubleshoot a slow SQL query?

Answer: Check the execution plan, row estimates, scans/seeks, joins, indexes, filters, statistics, and actual runtime behavior.

Q19. What is a primary key?

Answer: A primary key uniquely identifies each row and cannot contain NULL values.

Q20. What is a foreign key?

Answer: A foreign key references a key in another table and helps enforce referential integrity.

Q21. What is a surrogate key?

Answer: A generated key, often an integer or GUID, used as a stable identifier independent of business attributes.

Q22. What is a natural key?

Answer: A key derived from business data, such as a customer number, that naturally identifies an entity.

Q23. DELETE vs TRUNCATE?

Answer: DELETE removes selected rows and can use a WHERE clause; TRUNCATE removes all rows more efficiently in supported systems and has different logging/transaction behavior.

Q24. What is an aggregate function?

Answer: Functions such as COUNT, SUM, AVG, MIN, and MAX summarize multiple rows.

Q25. How do you find customers with no orders?

Answer: Use a LEFT JOIN from Customers to Orders and filter where the order key is NULL, or use NOT EXISTS.

Q26. How do you get top N per group?

Answer: Use ROW_NUMBER/RANK/DENSE_RANK partitioned by the grouping column and filter the desired rank.

Q27. What is EXISTS?

Answer: EXISTS tests whether a subquery returns at least one row.

Q28. What is a correlated subquery?

Answer: A subquery that references columns from the outer query and is logically evaluated in relation to each outer row.

Q29. Why avoid SELECT * in production queries?

Answer: It can read unnecessary columns, increase I/O, break consumers when schemas change, and reduce clarity.

Q30. What is NULL?

Answer: NULL represents an unknown or missing value; it is not equal to zero or an empty string.

Q31. How do you compare NULL?

Answer: Use IS NULL or IS NOT NULL rather than = NULL.

Q32. What is CASE?

Answer: CASE provides conditional logic in SQL expressions.

Q33. What is a transaction?

Answer: A transaction groups operations into a logical unit that can be committed or rolled back.

Q34. What is deadlock?

Answer: A deadlock occurs when transactions wait on each other's locks so neither can proceed.

Q35. How can you reduce deadlocks?

Answer: Keep transactions short, access objects in a consistent order, avoid unnecessary locks, and optimize queries.

Q36. What is query execution plan?

Answer: It shows how the database engine intends to access data and execute joins, filters, sorts, and other operations.

Q37. What is partitioning in a database?

Answer: It divides a large table into manageable physical/logical portions, often by date or range, to improve manageability and performance.

Q38. What SQL skills should a Data Engineer master?

Answer: Joins, aggregations, CTEs, window functions, deduplication, incremental logic, date handling, complex transformations, and performance troubleshooting.

Revision checklist: ■ I understand the concept ■ I can explain it without notes ■ I can give a practical example ■ I can answer follow-up questions

LEVEL 5 — DATABRICKS + PYSPARK

Question count: 37

Q1. What is Apache Spark?

Answer: Spark is a distributed processing engine designed to process large datasets across multiple machines.

Q2. What is Databricks?

Answer: Databricks is a cloud data and AI platform built around Apache Spark, with collaborative notebooks, workflows, governance, and lakehouse capabilities.

Q3. What is a Spark DataFrame?

Answer: A distributed collection of data organized into named columns, similar conceptually to a table.

Q4. Transformation vs action?

Answer: Transformations define a new dataset lazily; actions trigger execution and return or write results.

Q5. What is lazy evaluation?

Answer: Spark delays execution of transformations until an action requires the result, allowing it to optimize the execution plan.

Q6. Narrow vs wide transformation?

Answer: Narrow transformations generally avoid data movement across partitions; wide transformations require shuffle between partitions.

Q7. What is a shuffle in Spark?

Answer: Shuffle redistributes records across partitions, often for joins, groupBy, and aggregations.

Q8. Why is shuffle expensive?

Answer: It can involve network transfer, disk I/O, serialization, and additional coordination.

Q9. What is repartition?

Answer: It reshuffles data to create a requested number of partitions and can change distribution.

Q10. What is coalesce?

Answer: It reduces the number of partitions with less shuffling in common cases and is often used before writing fewer output files.

Q11. What is caching?

Answer: Caching keeps a dataset in memory or another storage level so repeated actions can reuse it.

Q12. When should you cache?

Answer: When the same expensive dataset is reused multiple times and caching provides a measurable benefit.

Q13. What is a broadcast join in Spark?

Answer: A small table is broadcast to workers so the large table can join locally, reducing shuffle when the size is suitable.

Q14. What is Spark SQL?

Answer: A SQL interface for querying structured data using Spark's distributed execution engine.

Q15. What is Delta Lake?

Answer: A storage layer providing transactional reliability and table-management capabilities on a data lake.

Q16. Why use Delta MERGE?

Answer: MERGE supports efficient upsert logic for inserts, updates, and sometimes deletes based on match conditions.

Q17. What is Delta time travel?

Answer: It allows querying or restoring earlier versions of a Delta table, subject to retention and configuration.

Q18. What is schema enforcement in Delta?

Answer: It helps prevent incompatible data from being written to a table.

Q19. What is schema evolution?

Answer: It allows supported schema changes to be incorporated under controlled conditions.

Q20. What is a notebook?

Answer: An interactive environment containing code, SQL, documentation, and outputs used for development and analysis.

Q21. What is a Databricks job/workflow?

Answer: It orchestrates notebook, Python, SQL, or other tasks on a schedule or trigger.

Q22. How do you optimize a Spark job?

Answer: Reduce unnecessary data, filter early, select needed columns, optimize joins, manage partitions, avoid excessive shuffles, and use appropriate file formats.

Q23. Why prefer Parquet/Delta over CSV?

Answer: They are typed, columnar, compressed, and generally more efficient for analytical processing.

Q24. What is a partition in Spark?

Answer: A partition is a chunk of distributed data processed by a task on a worker.

Q25. What is data skew in Spark?

Answer: Uneven distribution of records causes some tasks to process much more data than others.

Q26. How can you handle skew?

Answer: Use better partitioning, salting for extreme keys, broadcast joins for small sides, and Spark features suitable to the workload.

Q27. What is an action example?

Answer: count(), collect(), show(), and write operations trigger computation.

Q28. What is a transformation example?

Answer: select(), filter(), withColumn(), join(), and groupBy transformations build a new logical plan.

Q29. Why avoid collect() on large data?

Answer: It moves all results to the driver and can cause memory exhaustion.

Q30. What is the driver?

Answer: The driver coordinates the Spark application and creates/manages the execution plan.

Q31. What are executors?

Answer: Executors run tasks on worker nodes and store intermediate/cached data.

Q32. What is a cluster?

Answer: A set of compute resources used to run distributed Spark workloads.

Q33. What is the driver memory problem?

Answer: Large data collected to the driver or excessive driver-side operations can cause out-of-memory failures.

Q34. What is the medallion pattern in Databricks?

Answer: Bronze ingests raw data, Silver cleans/conforms it, and Gold provides business-ready datasets.

Q35. How do you handle bad records?

Answer: Validate, quarantine invalid records, log reasons, and continue valid processing when business requirements permit.

Q36. How do you make a Spark pipeline restartable?

Answer: Use deterministic inputs, checkpoints where appropriate, idempotent writes, batch IDs, and clear state management.

Q37. What PySpark skills should a Data Engineer know?

Answer: DataFrame operations, joins, aggregations, windows, UDF awareness, partitioning, file formats, Delta operations, and performance basics.

Revision checklist: ■ I understand the concept ■ I can explain it without notes ■ I can give a practical example ■ I can answer follow-up questions

LEVEL 6 — REAL PROJECT + SCENARIO INTERVIEW

Question count: 35

Q1. Explain your end-to-end Azure data project.

Answer: Explain source systems, ADF ingestion, ADLS layers, transformation using Databricks/Synapse, target warehouse, orchestration, monitoring, security, and CI/CD.

Q2. Why did you choose ADF?

Answer: ADF provides managed cloud orchestration, connectors, scheduling, parameterization, monitoring, and integration with Azure data services.

Q3. Why ADLS Gen2?

Answer: It provides scalable, cost-effective lake storage with hierarchical namespaces and security features suitable for analytics.

Q4. How would you process 100 files arriving daily?

Answer: Use event/schedule ingestion, metadata-driven configuration, Get Metadata or file listing, ForEach with controlled parallelism, logging, and processed-file tracking.

Q5. How would you process only new files?

Answer: Track file name/path, size/hash or arrival metadata in a control table and process only files not successfully recorded as completed.

Q6. What if the same file arrives twice?

Answer: Use idempotent processing and a unique file/batch key so the second arrival does not create duplicate target data.

Q7. What if one file fails in a batch?

Answer: Log the failed file, isolate it, continue other independent files where appropriate, alert support, and provide a safe reprocessing path.

Q8. What if ADF succeeds but the target has incomplete data?

Answer: Use reconciliation checks such as row counts, totals, keys, and batch status; investigate the target transaction/write and source extraction.

Q9. How would you implement incremental loading from SQL?

Answer: Use a reliable modified timestamp, increasing key, CDC, or change tracking; store a checkpoint and advance it only after successful processing.

Q10. How would you implement SCD Type 2?

Answer: Compare incoming and current dimension rows, expire changed current records, insert new versions with effective dates/current flag, and retain unchanged records.

Q11. How would you handle deletes?

Answer: Use CDC/change tracking when available, or source deletion indicators/snapshots, and apply a controlled delete or soft-delete strategy.

Q12. How would you handle schema changes?

Answer: Detect changes, validate compatibility, use controlled schema evolution where supported, and alert when breaking changes occur.

Q13. How would you troubleshoot a pipeline that suddenly became slow?

Answer: Compare current and historical runtimes, identify the slow activity, inspect data volume, source performance, partitions, parallelism, throttling, and transformation stages.

Q14. How would you optimize a large SQL extraction?

Answer: Filter incrementally, select only required columns, use indexed/change columns where appropriate, avoid expensive functions on filter columns, and validate the execution plan.

Q15. How would you optimize a large Spark transformation?

Answer: Filter/select early, use efficient formats, reduce shuffles, handle skew, choose sensible partitions, and avoid unnecessary actions.

Q16. How do you monitor production pipelines?

Answer: Use ADF monitoring plus centralized logs/alerts, track failures and SLA, and capture batch IDs, counts, duration, and data-quality metrics.

Q17. How do you secure production data pipelines?

Answer: Use managed identities, least-privilege RBAC/ACLs, Key Vault for secrets, encryption, private connectivity where required, and avoid hardcoded credentials.

Q18. How do you deploy ADF from Dev to Prod?

Answer: Use Git-based development and review, automated validation/build, environment-specific parameters, and controlled release to production.

Q19. How do you handle a failed production deployment?

Answer: Stop further promotion, inspect deployment errors, restore/redeploy the last known good version where appropriate, and document the incident.

Q20. How do you explain your role in a project?

Answer: State your responsibilities clearly: ingestion, pipeline development, transformations, SQL, monitoring, performance tuning, deployment, and production support.

Q21. Tell me about a challenging issue you solved.

Answer: Use Situation → Problem → Investigation → Action → Result. Quantify improvement when you genuinely know the numbers.

Q22. How do you answer when you don't know something?

Answer: Be honest: explain what you know, state the gap, and describe how you would investigate or learn it rather than inventing an answer.

Q23. What makes a data pipeline scalable?

Answer: Parallel processing, incremental ingestion, partitioning, efficient formats, reusable metadata-driven design, controlled compute, and minimal unnecessary data movement.

Q24. What makes a pipeline reliable?

Answer: Idempotency, checkpoints, retries, validation, monitoring, alerting, clear failure paths, and restartability.

Q25. What makes a pipeline maintainable?

Answer: Parameterization, reusable components, naming standards, metadata-driven configuration, documentation, logging, testing, and source control.

Q26. How do you perform data reconciliation?

Answer: Compare source and target row counts, sums, distinct keys, control totals, and batch-level status to identify mismatches.

Q27. What is a data quality framework?

Answer: A repeatable set of checks for completeness, validity, uniqueness, consistency, accuracy where measurable, and timeliness.

Q28. How would you design a generic ingestion framework?

Answer: Store source/target/configuration metadata, create reusable ingestion logic, parameterize connections/paths, log each batch, and handle full/incremental modes.

Q29. How would you design a multi-environment solution?

Answer: Keep code common, externalize environment-specific settings, use separate resources/data boundaries, and deploy through CI/CD.

Q30. How would you reduce pipeline cost?

Answer: Use incremental loads, avoid repeated full scans, right-size compute, optimize file sizes, reduce unnecessary transformations, and schedule workloads efficiently.

Q31. How would you handle sensitive data?

Answer: Classify it, restrict access, encrypt it, avoid unnecessary copies, use managed identity/Key Vault, and apply masking or tokenization where required.

Q32. How do you test a data pipeline?

Answer: Unit-test transformations, test positive/negative data cases, validate counts and business rules, test failure/retry paths, and perform end-to-end testing.

Q33. What is production support for a Data Engineer?

Answer: Monitoring jobs, troubleshooting failures, resolving data issues, rerunning safely, investigating performance, communicating impact, and preventing recurrence.

Q34. How do you describe an experienced Data Engineer mindset?

Answer: Think beyond building activities: design for scale, reliability, security, cost, observability, reusability, and operational support.

Q35. What should you avoid in a 4–5 year interview?

Answer: Do not claim tools or responsibilities you cannot explain. Interviewers often ask deep follow-ups, so every stated skill should be understood.

Revision checklist: ■ I understand the concept ■ I can explain it without notes ■ I can give a practical example ■ I can answer follow-up questions

12-Week Study Plan

Weeks	Study
1–2	Level 1 — Foundation: 36 Q&A + basic ADF/ADLS hands-on practice
3–5	Level 2 — Intermediate: ADF activities, parameters, dynamic pipelines, incremental patterns
6–7	Level 3 — Advanced: architecture, SCD, CDC, performance, security, CI/CD
8–9	Level 4 — SQL: 5 problems/day + explain every solution aloud
10–11	Level 5 — Databricks/PySpark: DataFrames, Spark, Delta, optimization
12	Level 6 — Project/Scenario: mock interviews, project explanation, troubleshooting

Important interview principle: Preparing for a 4–5 year interview level does not mean pretending to have 4–5 years of experience. Be truthful about your actual experience, while developing the technical understanding expected at that level.

Target outcome: You should be able to explain Azure Data Engineering concepts clearly, build and troubleshoot ADF pipelines, write strong SQL, understand Databricks/PySpark, and discuss production scenarios confidently.